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Analysis and optimization of energy flow in the full chain of carbon dioxide capture and oil recovery

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Abstract

CO₂ capture is a process with a high energy consumption, and its large-scale implementation should be based on comprehensive analysis of its impact on the energy, economy, and environment. The process of injecting CO₂ into existing oil fields is a well-known enhanced oil recovery (CO₂-EOR) technique. Using CO₂ as a working fluid to recover oil can compensate for the energy consumption of the capture and transport processes, increasing the feasibility of CO₂ capture while achieving carbon sequestration. In this study, a full-chain CO₂ capture, utilization, and storage (CCUS) system based on the post-combustion capture method is deconstructed and coupled. A full-chain energy consumption calculation software is developed, and optimization analysis of the energy consumption system is conducted. The energy budget of the oil displacement utilization is deconstructed, and the advantages of the water alternating gas (WAG) method are clarified from an energy budget point of view. The analysis reveals that the benefits of CO₂-EOR are far greater than the energy consumption of other CCUS processes, and CCUS-EOR is a CO₂ utilization method with positive energy benefits. Based on the simulation of the effects of N₂ and CH₄ on the recovery factor, a multi-well combined injection-production method is proposed, and the reasons for increasing profit are analyzed.

Highlights

Optimization analysis of the energy consumption for CCUS system is conducted.

The advantages of the WAG method are clarified from an energy budget point of view.

The CCUS-EOR is a CO₂ utilization method with positive energy benefits.

Keywords: Energy budget, Full-chain, Carbon dioxide capture, utilization, and storage (CCUS), CO₂ enhanced oil recovery (CO₂-EOR)

1 Introduction

The Paris Agreement's goal is to keep the increase in the global average temperature to well below 2 °C above pre-industrial levels. Therefore, the global power sector is expected to meet the rising demand as access to electricity increases and to provide for a low-carbon future

in which end-use activities are increasingly electricity-dependent. In the International Energy Agency's (IEA) Sustainable Development Scenario, in which global CO₂ emissions from the energy sector decrease to zero on a net basis by 2070, CO₂ capture, utilization, and storage (CCUS) accounts for nearly 15% of the cumulative reduction in emissions compared with the Stated Policy's Scenario [7]. The Global Carbon Capture and Storage Institute (GCCSI) has predicted that to achieve the Paris Agreement's target, global annual carbon capture must reach 5 billion tons by 2050, and more than 2732

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large-scale CCUS facilities will be required. This means that between 2020 and 2050, 90 CCUS facilities will be put into operation every year. According to *The Role of CCUS in Low-Carbon Power Generation Systems* report issued by IEA, 20 CCUS power generation projects are currently under development. Eleven of these 20 projects are in the United States; 3 are in China; 3 are located in the United Kingdom, and Ireland, Korea, and the Netherlands each have 1 project. The two large-scale CCUS power projects that are currently operational and the 20 in development have a combined potential capture capacity of more than 50 Mt of CO₂ per year. This is lower than the stated 310 Mt CO₂ that will need to be captured from power generation in 2030 under the IEA's Sustainable Development Scenario, reflecting that carbon capture, utilization, and storage related to power generation are not currently on course.

The application and promotion of CCUS requires a synergy of policies, capital, and technology. At the technical level, CO₂ source-sink matching, the total cost of the full CCUS chain, and the reduction of the total CO₂ emissions are issues that need to be comprehensively considered [18].

Regarding source-sink matching, Al-Mohannadi and Linke [1] proposed an approach to the systematic design of low cost carbon integration networks for industrial parks through integrated analysis of the sources, utilization, and storage options, as well as the capture, separation, compression, and transmission options. Tan et al. [17] and Lee and Hash [11] developed a source-sink matching model, which provides a methodological basis for the rational use of source and sink resources. Fan et al. [5] developed an optimization model that does not consider the CCUS cost constraints in determining the CO₂ reduction potential of existing coal-fired power plants with CCUS systems in China from the perspective of source-sink matching. Ecofys developed a series of methodologies to calculate the underground storage potential of each type of reservoir [4].

Regarding the cost of the CCUS chain, many studies have focused on the specific technical and economic evaluations of CCUS. CCUS demonstration projects launched in China have demonstrated that economic feasibility has become a substitute for technical feasibility and is the main obstacle to the large-scale industrialization of CCUS technology [21]. Zhang et al. [24] evaluated the investment in CCS (carbon capture and storage) retrofitting for two typical types of power plants from the perspective of power generation enterprises. Their results indicate that a large gap exists between the carbon price needed for CCS retrofitting of both typical types of power plants and the current prices on the voluntary emission reduction market. Selosse and Ricci [15]

investigated various scenarios with different carbon storage potentials and examined the impact of an increase in carbon transport costs on CCS development. Durmaz [3] employed a simple energy generation model, scrutinized the economic drivers of CCS based on the analytical results, and discussed the possible obstacles that could prevent the widespread implementation of the technology. Lin and Tan [13] proposed a deferred option model to quantitatively evaluate the impacts of low oil price shocks and the carbon trading mechanism on CCUS-enhanced oil recovery (EOR) project investments based on real options theory. They found that the current economic environment is not sufficient to trigger immediate investment in CCUS projects, but the introduction of a carbon trading mechanism could significantly improve the investment value of such a project and reduce the probability of investment failure. As a value-adding benefit, CO₂ injected into a mature oil reservoir can enable incremental oil to be recovered. Typically, CO₂ flooding allows the recovery of an additional 10–15% of the original oil in place (OOIP) (2010). CO₂-EOR is currently one technique being considered as an ultimate long-term geological storage solution for CO₂ owing to its economic profitability, i.e., the incremental oil production offsets the cost of the carbon sequestration [19]. Fleten et al. [6] used a traditional dynamic programming approach to investigate the profitability of several value chains for large CO₂ producers. Their main conclusion is that CO₂-EOR will result in a situation more profitable than venting the CO₂ or investing in CCS alone. The oil revenues from EOR are the main driver of CCS+EOR implementation.

The economic evaluation of the full-chain is greatly affected by energy prices, interest rates, carbon taxes, and renewable energy compensation policies, and it is time-sensitive, whereas energy budgets are not affected by the above factors and are deterministic. In this study, CCUS-EOR based on the post-combustion capture method was analyzed, and a full-chain energy consumption calculation software was developed. By analyzing the energy budget of the CO₂ flooding system, the differences in the energy budgets of different injection designs were investigated, and a flooding scheme that improves the energy yield was determined.

2 Deconstruction and coupling of the full-chain system

The main processes of the full-chain system include CO₂ capture, transport, and utilization. To meet the conditions for transport and utilization, compression, and purification processes need to be applied after capture. These processes are generally conducted directly in the power plant. In this study, the power plant, CO₂

capture, CO₂ compression, and purification were collectively referred to as the source of the CO₂. During the oil displacement, to ensure continuous operation, CO₂ buffer facilities are necessary, so a storage subsystem should be included in the full chain. When the storage pressure is set, the parameter passing of the pressure between the injection and the transport is blocked, and the complexity of the calculation is reduced. In this way, the full-chain CCUS-EOR system was deconstructed and coupled into five subsystems: CO₂ source, transport, storage, oil displacement, and separation. Figure 1 shows the key parameters and their passing in each subsystem of the full-chain CCUS-EOR system. The subsystems are thermodynamically connected by the parameter passing, including the mass flow rate, pressure, temperature, and fluid composition and content.

In the full-chain system, the oil displacement process is regarded as a quasi-closed cycle, and the flow includes CO₂ injection, CO₂ oil displacement in the oil reservoir, CO₂ production, and separation. During the operation of the system, some CO₂ leaks out, some is captured by the oil reservoir, and some is not separated from the associated gas. The separated CO₂ and supplementary CO₂ from the storage system are compressed to the injection pressure and are reinjected. The other processes in the full-chain system are regarded as processes with a constant flow of CO₂.

For the oil displacement subsystem, to achieve the miscible pressure, the CO₂ injection pressure is usually relatively high. In this study, considering the safety of transport, the pressure of the CO₂ in the storage

subsystem was set to 10 MPa. The final compression pressure at the power plant (referred to as the transport pressure) is determined by the transportation conditions and storage pressure.

3 Subsystem modeling and optimization

3.1 CO₂ source

The power consumption of capture system is defined as E_{1-1} , and.

$$E_{1-1} = (\eta_l, \text{capture process and technology}, x_{co2}, y_{impure}), \quad (1)$$

where η is the thermal efficiency of the power station, which depends on the power plant's processes and technology; x_{co2} is the CO₂ content; and y_{impure} is the content and composition of the impurities.

The simulation optimization of the CO₂ capture system revealed that there are two ways to reduce the power consumption.

- (1) Improve the efficiency of the power station. When the thermal efficiency of the power station is improved, the coal consumption of the system with a fixed net output decreases, the amount of CO₂ that needs to be captured decreases, the total energy consumption for capture decreases, and the net CO₂ emissions decrease.
- (2) Reduce the equivalent work consumption of the capture system. For the post-combustion capture system, the energy consumption mainly occurs in the desorption process, and the absolute value

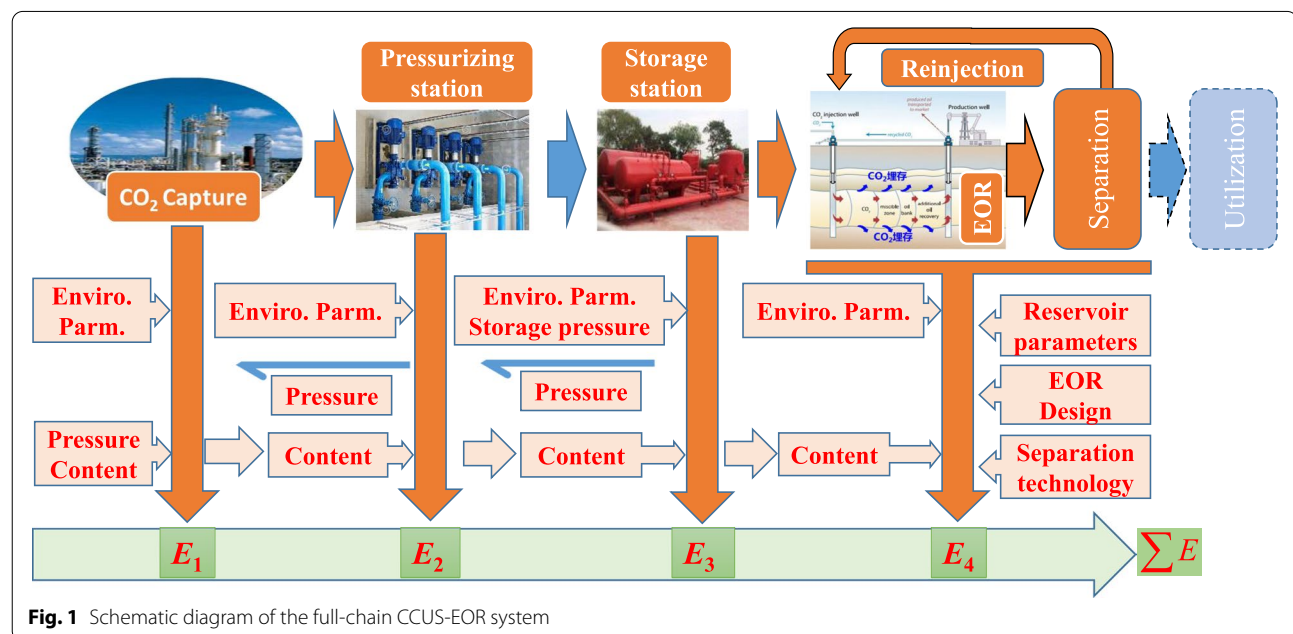


Fig. 1 Schematic diagram of the full-chain CCUS-EOR system

of the heat consumption or the reduction of the energy grade requires the development of new or improved existing absorbents. In addition, the use of new low-energy capture methods, such as pressurized oxygen-enriched combustion and chemical chain combustion, is also an effective way to reduce the energy consumption.

In this study, the reference plant was an air-based supercritical pulverized coal Rankine cycle plant with a net output of 550 MW. The post-combustion CO₂ recovery technology was designed to remove 90% of the CO₂ from the flue gas based on a compromise between energy and CO₂ emissions [12]. An monoethanolamine (MEA)-based absorption system was used to capture the CO₂, which has been demonstrated to be one of the most mature and least expensive technologies. For an amine CO₂ capture system, a large amount of heat is required to regenerate the solvent in the stripper. This heat is provided by the steam drawn from the steam cycle. Compressing the captured CO₂ for pipeline transport also consumes substantial electrical energy, which is also provided by the plant. To ensure a net output of 550 MW, the steam turbine needs to generate more power and the boiler needs to burn more coal. In this study, it was assumed that the parameters (temperature and pressure) of the steam extracted from the steam turbine can be changed according to the demand of the stripper in the capture system without affecting the performance of the steam turbine itself.

The thermal efficiency (η_t) of the plant was set to 40.62%, which is the average for 113 supercritical wet cooling units (600–660 MW) in China.

The heat requirements for solvent regeneration mainly depend on the lean sorbent and other system parameters. This heat is provided by the low-pressure steam extracted from the steam turbine. Rochelle [14] summarized the evolution of the heat load for post-combustion CO₂ capture since 1976, and the reboiler load decreased from nearly 12 MJ/kg CO₂ in 1978 to less than 4.3 MJ/kg CO₂. In 2009, for the plant in Mitsubishi, the heat consumption is expected to decrease to less than 2.5 MJ/kg CO₂ [8]. The first commercialized CO₂ capture plant built by Shell reduced the capture heat consumption to 2.33 GJ/t CO₂ [16]. Experiments [20] have shown that for a MEA solvent system with a mass fraction of 20%, the optimal energy consumption for CO₂ regeneration is about 3.6 GJ/t CO₂. In this study, considering the technological maturity and development of the existing technology, the heat consumption was set to 2500 kJ/kg CO₂. In addition, the circulation of MEA solution in the capture system, the flow of the flue gas, and the discharge

of the waste heat all require energy to run the pumps, fans, and other equipment. According to Desideri and Paolucci [2], the power consumption of the post-combustion capture system is 13–120 kWh/t CO₂ (46.8–432 kJ/kg CO₂). Based on this and the simulation calculation of the capture system, the power consumption was assumed to be 135 kJ/kg CO₂.

Except for CO₂, most of the other impurities in the captured gas are non-condensable gases, such as N₂, O₂, and Ar, which will affect the power consumption of the compression process in preparation for transport and injection and may have an impact on the oil displacement. When the MEA-based absorption method is used to remove CO₂ from the flue gas, the separated CO₂ content can exceed 98%. In this study, the CO₂ content (x_{CO_2}) was assumed to be 98.6%. Owing to a lack of research results on the impact of the above-mentioned impurities on oil displacement and, the impurities were attributed to N₂ in the simulation, that is, the captured gas was assumed to contain only two components, CO₂ and N₂, and the other impurities were calculated as N₂. For the transport system, replacing other impurities with N₂ will cause a change in power consumption, but calculations show that the error is less than 0.5%.

The power consumption of the compression system is E_{1-2} , and,

$$E_{1-2} = f(t_0, p_1, \text{capture process and technology}, p_2, x_{CO_2}, y_{\text{impure}}), \quad (2)$$

where t_0 is the ambient temperature; p_1 is the pressure of the captured gas at the outlet of the capture system; and p_2 is the transportation pressure.

For the MEA-based absorption method, the purification of the captured CO₂ with a high purity is not necessary. Drying and compression are conducted simultaneously.

Multi-stage compression and intermediate cooling were adopted, and the adiabatic efficiency of the compressor was set to 80%. Based on experience, the temperature of the discharged gas at the outlet of the compressor does not exceed 150 °C, and the compression ratio is about 3, so the optimum intermediate pressure can be determined. The compression system technology is mature, and optimization is quite unlikely.

3.2 Transportation

For large-scale application, pipeline transport is considered. Since the factors influencing the energy consumption related to transportation are clear, to simplify the calculations, the transport path was set as land and plains, with a 45° elbow every 10 km. The transport length was set to 100 km.

The resistance loss was calculated as follows:

$$dp = \rho f \frac{dL}{D_i} \frac{v^2}{2}, \quad (3)$$

where ρ is the density; f is the Fanning friction factor; L is the distance/depth; D is the well diameter and v is the flow rate.

The energy consumption related to transportation is the power consumption of the compression in the booster station:

$$E_2 = f \left(t_0, \text{compression process and technology}, x_{\text{CO}_2}, y_{\text{impure}}, p_2, p_3, \text{pipeline parameters} \right), \quad (4)$$

where p_3 is the pressure in the storage subsystem.

3.3 Injection and production

When CO_2 is injected into a wellbore, there is friction and heat transfer occurs between the borehole wall and the fluid. The calculation model has been described by Zhang et al. [22], and Jiang et al. [9]. The oil displacement injection pipe (DN65) has an inner pipe diameter of 67 mm and an outer pipe diameter of 76 mm.

The power consumption related to the injection includes three compression processes: compressing the CO_2 separated from the produced gas and supplemented from the storage system to the injection pressure; and compressing the water to the injection pressure, if the Water Alternate Gas (WAG) flooding method is used.

The power consumption related to the production wells includes the work required to suck out the gas and liquids, i.e., the crude oil and water.

When writing the program, the bottom hole pressure was optimized via EOR modeling, and then, the injection pressure and power consumption were calculated.

3.4 CO_2 -EOR flooding

The Petrel software was used to simulate and optimize the oil displacement. Oil recovery is influenced by the geologic, structural, and stratigraphic features. To clarify the influence of the flooding method on the energy budget of the full-chain, a typical reservoir model was designed. A 1:1 well-pattern configuration (one injection well and one production well) was adopted. The reservoir depth was set to 800–810 m, the effective thickness of the oil layer was 10 m, the oil reserve was $3.35 \times 10^4 \text{ m}^3$, and the original pressure of the reservoir was 20 MPa. The purpose of this model design was to minimize the influence of the reservoir itself to focus on the influence of the EOR flooding method and the gas injection parameters.

3.5 Produced gas separation

Oil displacement in a typical reservoir was simulated. Figure 2 shows the CO_2 and CH_4 contents of the produced gas for three different impurity contents. The composition of the produced gas was analyzed, and it was found that in the initial phase of injection, the gas produced only contained the gas from the crude oil. During the gas injection, CO_2 breakthrough occurred, CO_2 began to appear in the produced gas, and its content continued to increase.

For pure CO_2 injection, it was found that the produced gas did not contain CO_2 until 3.13 years after the CO_2

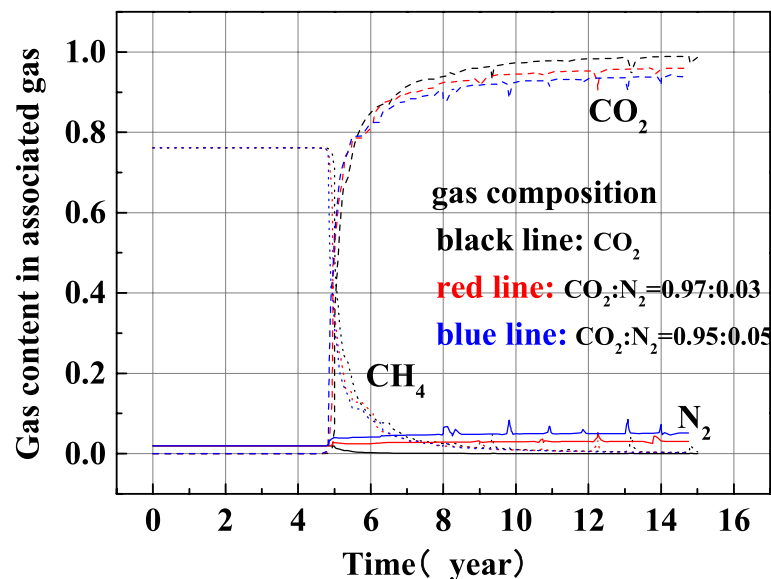


Fig. 2 Variations in the N_2 , CO_2 , and CH_4 contents of the produced gas with injection time

injection, and the methane and ethane contents of the produced gas reached 94.8% (76.09% + 18.66%) or greater. From 3.13 years to 4.93 years, the CO₂ content was 1%. After 4.93 years, the CO₂ content increased rapidly, and it only took 26 days to increase from 1 to 28%. During the next 134 days, the CO₂ content increased from 28 to 70%, and it reached 90% after 529 days. Finally, the average CO₂ content remained above 96%. The simulation results indicate that the proportion of CO₂ kept changing throughout the production period. Large changes in the associated gas volume and the CO₂ content severely affect the operation of the equipment, especially the gas processing facilities. Therefore, once gas breakthrough occurs, the produced gas must be separated for re-injection to avoid excessive purchase of CO₂ and greenhouse gas emissions. This requires a huge set of gas processing equipment. According to research and practice [23], the following treatment processes can be used for different CO₂ gas contents.

- i) When the CO₂ content of the produced gas is 97% to 98%, the produced gas often does not require decarbonization treatment. After pressure regulation, it is directly reinjected into the reservoir for CO₂ flooding.
- ii) When the CO₂ content of the produced gas is less than 3%, and the methane content is greater than 92%, it is close to the technical indicators for commercial natural gas, so it does not need to be decarbonized, and it can be used as commercial gas after water dew point control.

- iii) When the CO₂ content is between 3 and 97%, the CO₂ needs to be separated from the produced gas. The CO₂ separated from the production gas is reinjected, and the methane is used as commercial gas.

For the model used in this study, since the content of light hydrocarbons in the associated gas exceeded 94% in the first 4.93 years, it could be directly exported as commercial gas. Thereafter, the CO₂ content exceeded 95%, and it could be dried and pressure-regulated and then mixed with the supplementary gas to create the gas to be reinjected. For the following two years (days 1816–2479) due to the large changes in the CO₂ content, the CO₂ treatment process was relatively complicated.

The most promising high-CO₂ treatment methods are cryogenic distillation, triethanolamine (TEA) for large-scale CO₂ removal, and membrane permeation. Clifton Goodin, Gene Goar, and others have investigated the investment and operation energy consumption of the CO₂ separation units used in the EOR projects run by Shell, Monsabto, Union Texas, and Dow in many oil fields and refineries, and they calculated the relative cost of the gas treatment processes for CO₂ contents ranging from 20 to 90 mol%. Figure 3 shows the equivalent heat requirements (CO₂ outlet pressure of 3.1 MPa) for the four CO₂ removal methods.

For CO₂-EOR, the gas produced contains large amounts of methane and ethane, and the recovery of hydrocarbons is beneficial to reducing the energy consumption and improving the economy of the project. Thus, in this study, the cryogenic distillation method was used to extract the CO₂ from the produced gas, the

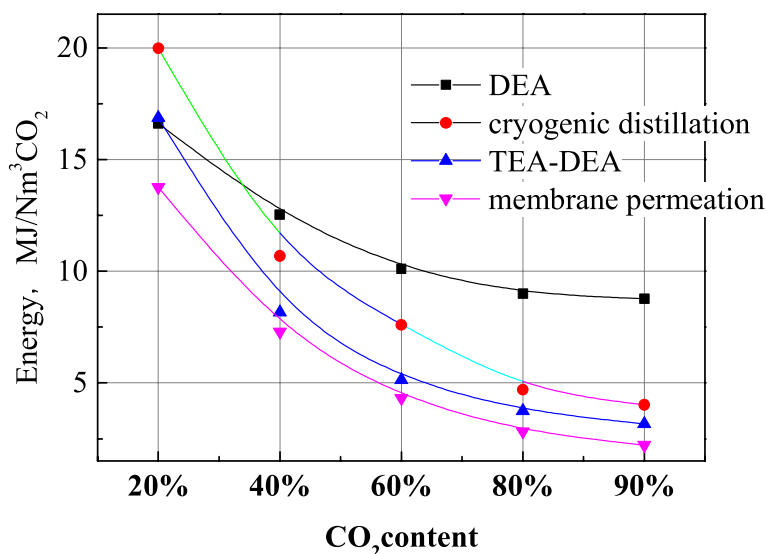


Fig. 3 Equivalent heat requirements for CO₂ extraction processes

separated CO₂ was reinjected, and the natural gas was sold.

Based on the above results, the energy requirement correlation was fitted for the cryogenic distillation method:

$$E_{sep} = -10.55 \ln(x) + 2.2972. \quad (5)$$

Then, the benefits of the oil displacement and separation process, E_4 , can be expressed as

$$E_4 = \sum (\text{Oil Equivalent Work} + \text{Associated Gas Equivalent Work} - \text{production gas separation work} - \text{injection and production work}). \quad (6)$$

The produced gas was condensed and the heavy hydrocarbons were separated. Then, the CO₂ was extracted. It was assumed that the calorific value of the mixed gas was 42 MJ/Nm³, which is same as that of the associated gas even though the mixed gas contained a small amount of N₂. For the energy balance calculations, the calorific value of the natural gas was set to 44 MJ/kg. The fuel heat energy and work equivalent value was set to 0.45.

4 Energy flow and carbon footprint of CO₂ flooding in the full-chain

With the help of the Petrel software, the CO₂-EOR process was optimized for the reservoir model with typical physical characteristics, and the energy flow and carbon footprint of the full-chain CCUS-EOR system were analyzed.

4.1 Influence of EOR methods

In this study, the simulation optimization was conducted for CO₂ injection into an oil reservoir via either as continuous gas injection (CGI) or WAG. The results confirm that WAG is an economical and feasible method. The results are shown in Table 1.

It can be seen from Table 1 that the WAG method greatly improves the benefits of EOR. The increase in the profit is mainly due to the following three aspects. i) The WAG method reduces the significant fluctuation of the

gas produced, delays the rate of increase of the gas-oil ratio, and increases the displacement life. The maximum limit of the gas-oil ratio was set to 3000 in this study; otherwise, the oil displacement process stopped operating. For continuous pure CO₂ injection, it took 3422 days to reach this limit, while it took 5764 days when the WAG method was used. ii) The WAG method reduces the output of associated gas, which in turn reduces the separation work. iii) Owing to the alternate injection of water,

the amount of CO₂ (sequestration) is greatly reduced, resulting in a significant increase in the unit CO₂ profit.

The WAG method greatly reduces the amount of CO₂ sequestration. In the case without considering the carbon tax, it is beneficial for oil companies to reduce the amount of CO₂ usage, thereby reducing the cost. When the environmental benefits of CO₂ sequestration are considered, the carbon tax benefits should be taken into account. Figure 4 shows the economic change in the CO₂-EOR of the simulated reservoir in response to changes in the carbon tax (only the oil and gas revenue and carbon tax are considered). The electricity price was set to 0.38¥/kWh. It can be seen that as the carbon tax increased, the economic advantage of the WAG method decreased comparing to the continuous pure CO₂ injection method. However, even when the carbon tax reached €55, the economic advantage of the WAG was still obvious.

4.2 Influence of injection gas composition

The continuous gas injection methods were simulated. The gas was CO₂ or CO₂ with different proportions of N₂. The CO₂ content of the gas captured using the post-combustion capture method is > 98%, while it is less than 98% for oxygen combustion capture method. The data analysis results are shown in Table 2.

Table 1 Energy analysis results of CO₂-EOR simulation for the typical reservoir model

EOR method	Production		Equivalent work consumption				CO ₂ sequestration	Equivalent work gained per kg CO ₂
	Oil	Associated gas	CO ₂ extraction	Gas injection	Water injection	Gas and water production		
	m ³	m ³	MJ	MJ	MJ	MJ	kg	MJ/kg CO ₂
CGI (CO ₂)	18,434	5,657,029	−6,742,346	−1,322,319	0	−104,616	16,729,850	17.6280
WAG (CO ₂)	21,045	4,274,791	−5,378,858	−822,605	−777,682	−167,775	8,001,782	42.094
Increase	13.57%		−20.22%	23.91%			−52.17%	138.78%

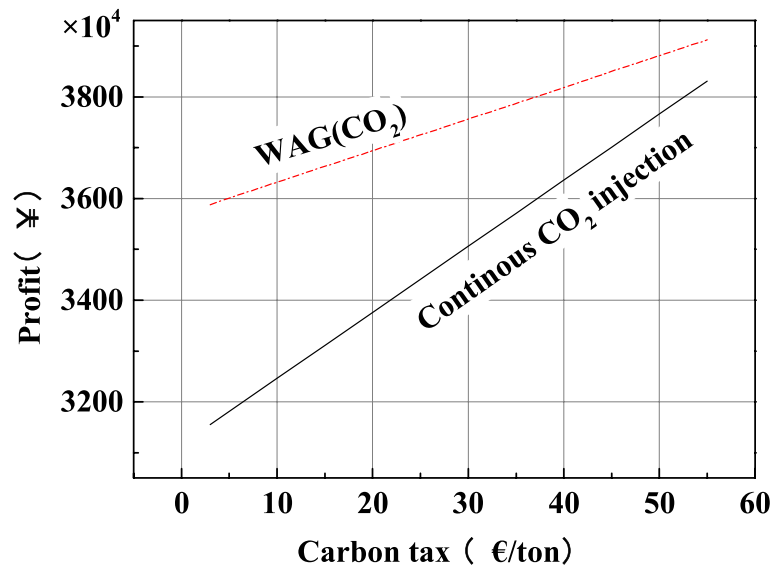


Fig. 4 Effect of the carbon tax on the economics of CO₂-EOR for the typical reservoir model

It was found that the difference in the cumulative oil was within 1% for the injection of gases with different N₂ contents. Considering the simulation error, it can be concluded that the N₂ content has no obvious effect on the oil yield within the simulated range. However, the unit CO₂ equivalent work gained increased with increasing N₂ content. This is mainly because the presence of N₂ reduced the CO₂ usage, but the oil production was only slightly different.

4.3 Influence of EOR methods

Crude oil contains light hydrocarbons such as CH₄, which are partially separated from the crude oil during the production process. Figure 5 shows the variations in the CO₂ and CH₄ contents of the gas produced when the CO₂ continuous injection and WAG methods are used.

It can be seen from the simulation in Sect. 4.2 that the presence of a small amount of non-condensable gas has little effect on the oil recovery. If the CH₄ content of the produced gas is less than 3% after CO₂ breakthrough occurs, the produced gas can be directly reinjected without separation. This reduced the separation work and improves the benefits of the full-chain system. Therefore, a flooding scheme with a variable-CO₂-content was designed, in which the CO₂ and CH₄ contents of the injected gas change with increasing injection time. The gas composition was designed based on the simulation results for continuous CO₂ injection. That is, when the average CO₂ content of the produced gas exceeded 96% and 98%, it was used as the gas injection composition. The results are shown in Table 3.

It can be seen from Fig. 5 that the WAG method increases the oil recovery factors because the intermittent water

Table 2 Simulation results for a reservoir with typical physical characteristics

Boundary conditions: The pressure at the bottom of the production well was set as 5 MPa							
Injection gas content	Production oil	Injection gas		Oil change ratio		EOR duration	Equivalent work gained
		Total gas	CO ₂				
	m ³	m ³	t	t _{oil} /t _{gas}	t _{oil} /t _{CO₂}	day	MJ/kg CO ₂
CO ₂	18,434	13,051,111	25,799	0.5674	0.5674	3422	17.628
CO ₂ :N ₂ =0.989:0.011	18,344	12,836,782	24,994	0.5741	0.5828	3377	18.552
CO ₂ :N ₂ =0.97:0.03	18,353	12,758,040	24,463	0.5844	0.5958	3351	18.174
CO ₂ :N ₂ =0.95:0.05	18,507	13,350,481	25,072	0.5674	0.5862	3468	18.509
CO ₂ :N ₂ =0.90:0.10	18,501	13,352,959	23,757	0.5780	0.6184	3468	19.487

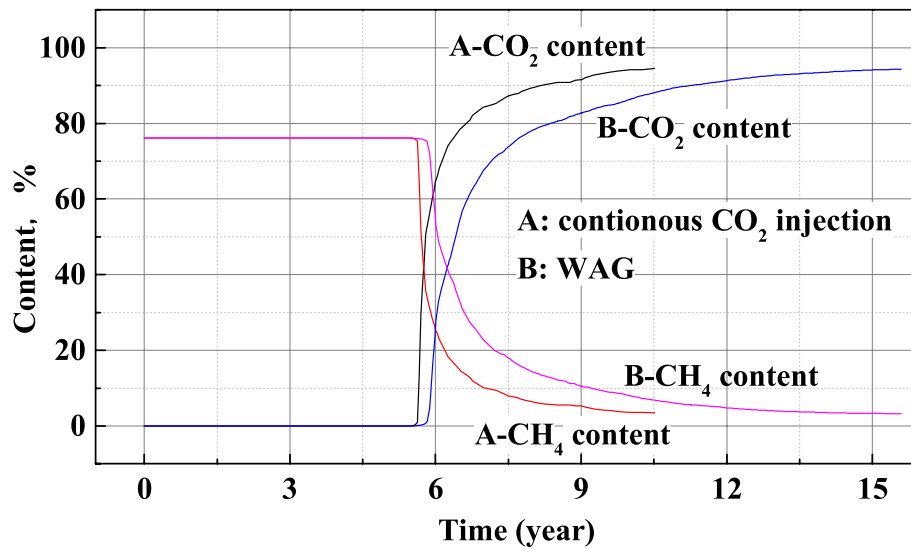


Fig. 5 Variations in the CO₂ and CH₄ contents of the produced gas with time

Table 3 Influence of the CH₄ contents of injected gas on the energy budget of EOR

perEOR method			Production		Equivalent work consumption		Equivalent work gained	CO ₂ sequestration	Equivalent work gained per kg CO ₂
			Oil	Gas	Gas separation	Injection and production			
			m ³	m ³			MJ	MJ	MJ
CGI (CO ₂)			18,434	5,657,029	−6,742,346	−1,426,935	295,040,745	16,729,850	17.628
WAG (CO ₂)			21,045	4,274,791	−5,378,858	−1,768,063	337,217,574	8,001,782	42.094
Mean injection gas content (CO ₂ : CH ₄)	0.98	CGI	19,677	6,255,996	−7,804,335	−1,621,566	316,460,138	16,342,603	19.355
		WAG	21,125	4,261,366	−5,588,987	−1,824,216	339,535,341	7,700,037	44.042
	0.96	CGI	19,455	5,788,515	−7,305,274	−1,598,413	313,101,857	15,476,283	20.221
		WAG	20,967	6,416,204	−8,769,794	−2,133,706	335,835,226	7,855,226	42.713

injection causes the flow of the produced gas to fluctuate less, effectively reducing the gas-oil ratio of the produced gas and increasing the EOR duration by more than 40%.

It can be seen from Table 3 that allowing the existence of CH₄ improves the income per unit CO₂ compared with injecting pure CO₂. This may be because CH₄ is a type of organic matter, which is soluble in crude oil, so its presence promotes fluidity and improves oil recovery. For the WAG method, although the power consumption of the injection and production processes increases, the separation work decreases more. The equivalent work gained for a mean CH₄ content of 98% is higher than that for 96%.

4.4 Multi-well combined injection and production EOR design

It can be seen from Fig. 5 that the CO₂ content of the produced gas quickly reached >95% after CO₂ breakthrough

occurred. After cooling, the condensable gas was separated. Except for the CO₂, most of the remaining gas was CH₄. Based on the results presented in Sect. 4.3, variable-component gas injection has no significant adverse effects on the unit CO₂ income. To decrease the separation work, we propose a direct reinjection scheme in which the produced gas is used as injection gas after cooling.

Figure 6 shows a schematic diagram of the design of the multi-well combined injection and production (MWCIP) scheme. The key to the realization of MWCIP is to set up a control system to dynamically monitor the composition of the gas produced by different production wells, and mix the gas separated from the production well with the supplementary gas of the buffer tank according to the designed injected gas composition. At the beginning of the EOR, CO₂ is injected,

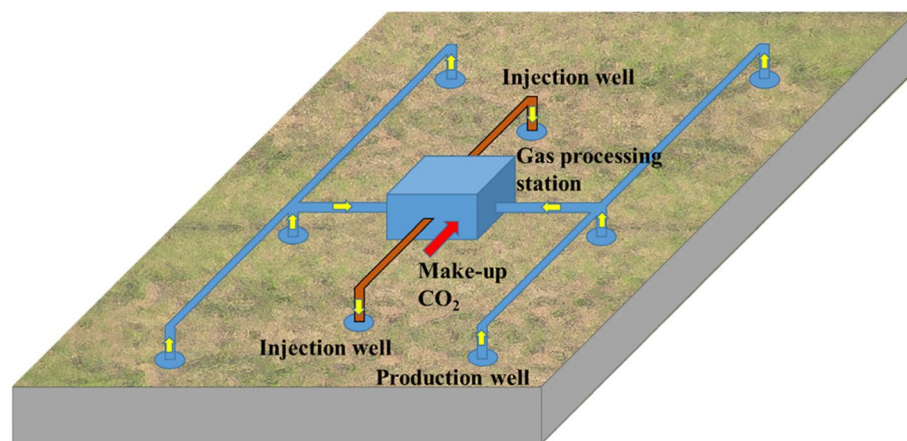


Fig. 6 Schematic diagram of the design of the multi-well combined injection and production (MWCIP) scheme

and the associated gas is composed of hydrocarbons. After cooling and other handling processes, the gas can be exported as natural gas. After a period of time, CO_2 begins to appear in the produced gas. When the CO_2 content does not exceed the allowable CO_2 content for natural gas, the associated gas does not need to be separated. When the content increases further, the produced gas needs to be separated, the separated light hydrocarbons are exported as natural gas, and the CO_2 is reinjected. When the CO_2 content of the produced gas exceeds the set value (e.g., 94%, the CO_2 content can exceed 96% when mixed with the supplementary CO_2 from the storage tank), then after the necessary treatment, the produced gas is mixed with supplementary CO_2 and is reinjected. The composition of the reinjected gas changes with the composition of the produced gas, which is referred to as variable composition injection.

For the WAG method, the gas injection into a single well is discontinuous, while the production is continuous. Simulation and application results have shown that the flow rate of the produced gas accounts for 38–54% of the injection flow rate. Thus, two or more injection-production modules are used for joint operation. When the CO_2 content of the production gas is greater than the set value, the production gas enters the processing station first. After the gas is cooled, complemented, mixed, compressed, and subjected to other processes, it is reinjected according to the injection plan.

The advantages of MWCIP is that after the CO_2 content reaches a high value, the produced gas can be used as reinjection gas after simple gas processes, such as cooling, which decreases the separation work.

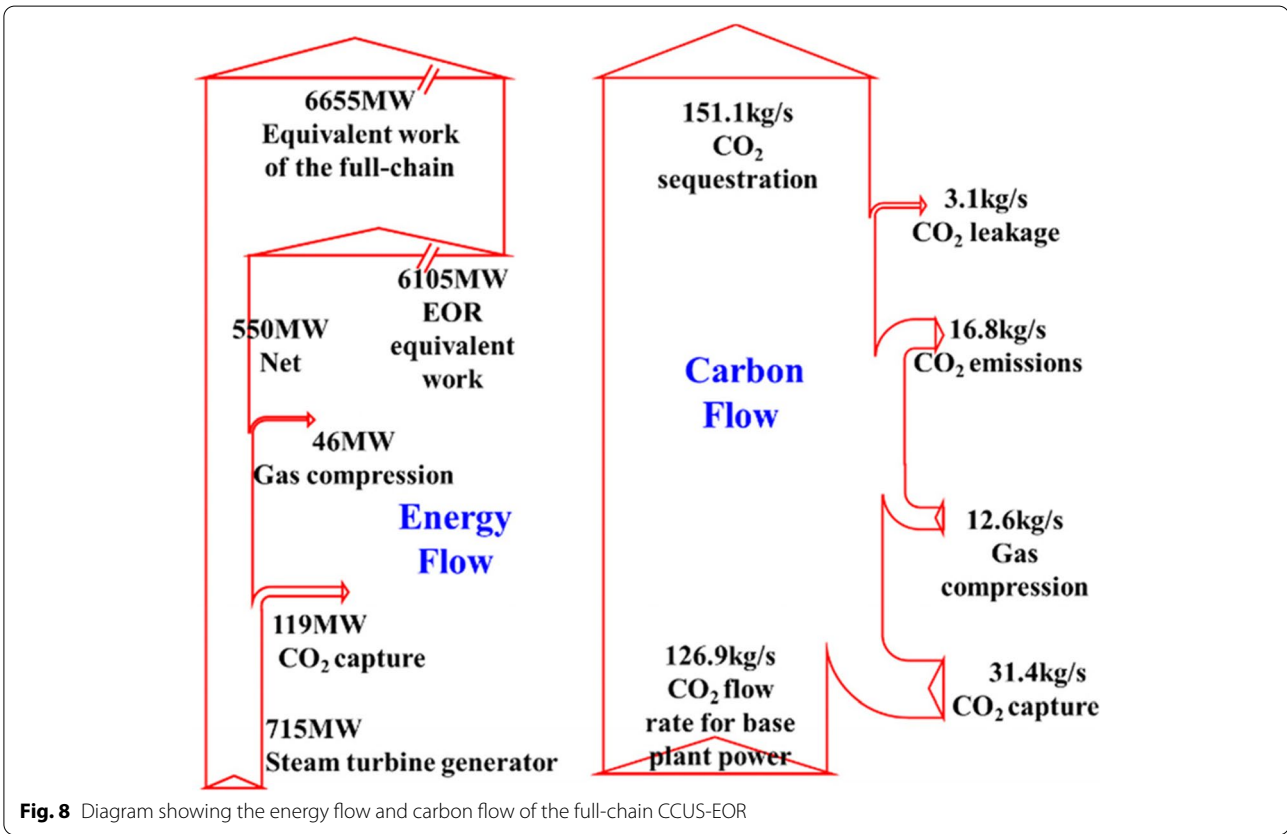
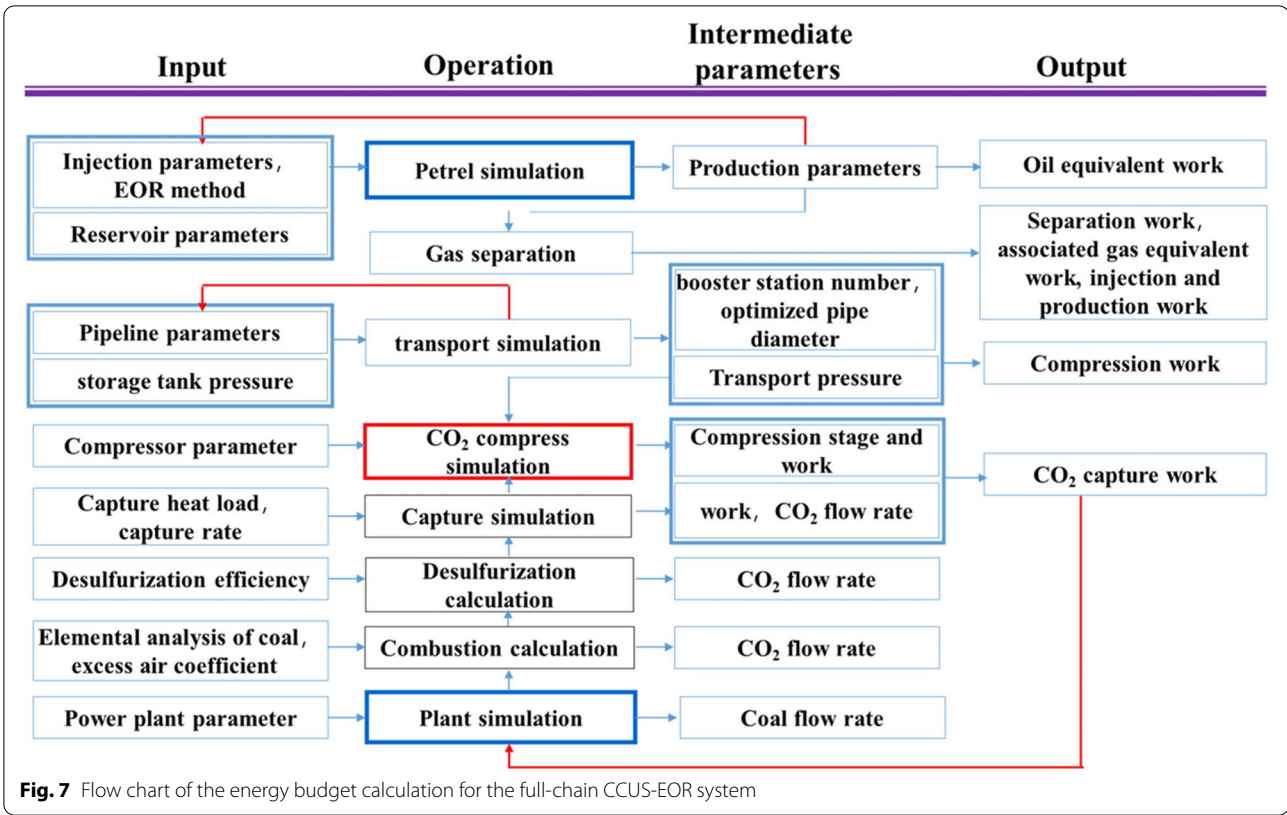
4.5 Full-chain CCUS-EOR energy flow and carbon footprint

A flow chart of the energy calculation for the full chain is shown in Fig. 7. The energy flow of the full chain and a carbon flow diagram of the oil displacement for the WAG method with variable composition gas injection (the average CO_2 content of the injected gas is 98%) are shown in Fig. 8.

It can be seen that the equivalent power consumption of the post-combustion capture and compression accounts for 30% of the net output of the power station based on the same net output power and a capture rate of 90%. In addition, it was found that the oil displacement equivalent energy gain is much higher than the power consumption of the capture, compression, and other processes, that is, CCUS-EOR is an application with a positive energy gain. It can be seen from the schematic diagram of the carbon flow that the actual reduction of the CO_2 emissions is 84% compared to a power station without CO_2 capture.

5 Conclusions

- (1) From the perspective of the energy budget, the energy gained in CCUS-EOR is greater than the energy consumed. CCUS-EOR is a CO_2 utilization application with positive energy benefits.
- (2) The energy budget of the oil displacement was deconstructed, and the advantages of the WAG method were confirmed. The source of this advantage is that the intermittent injection of water reduces the gas-oil ratio, increases the EOR duration, and improves the recovery factor. The flow rate of the produced gas decreases, which reduces the separation work. The amount of CO_2 used



decreases, and the equivalent work gained of per kg of CO₂ increases.

- (3) The presence of CH₄ in the injection gas can improve the recovery; therefore, a multi-well combined injection-production process is proposed. The reason why the MWCIG method improves the profit is that when the CO₂ content of the production gas exceeds a certain value, the associated gas is not separated and is reinjected directly after simple gas processing, which reduces the separation work, thereby further improving the benefits of the full-chain system.

Abbreviations

D: Well diameter (m); E: Power consumption (MW); f: Fanning friction factor; L: Distance/depth (m); p: Pressure (MPa); x: Gas content; y: Gas content; η_t : Thermal efficiency (%); ρ : Density (kg/m³); EGS: Enhanced geothermal system; EOR: Enhanced oil recovery; CCS: Carbon capture and storage; CCUS: CO₂ capture, utilization, and storage; CGI: Continuous gas injection; MEA: Monoethanolamine; MWCIP: Multi-well combined injection and production; WAG: Water alternating gas.

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Authors' contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by FZ, XR, HY, FX and JP. The first draft of the manuscript was written by FZ and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials

All data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript "Analysis and optimization of energy flow in the full-chain of carbon dioxide capture and oil recovery".

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